

Statistical and Machine Learning

Dimension reduction

Instructions

All questions should first be discussed in pairs or small groups. Afterwards, we shall discuss them together. Solutions will be distributed after each session.

Question 1 – PCA: encoding, decoding, and variance

PCA can be derived either by minimising the reconstruction error or by maximising the variance of the projected data.

- Explain in your own words what is meant by the “encode” and “decode” steps in PCA, and write down the reconstruction error $\mathcal{L}(\mathbf{W})$.
- What is the optimal choice of $\mathbf{W}$ that minimises this error, and how is it related to the empirical covariance matrix $\hat{\Sigma}$?
- Explain why minimising the reconstruction error is equivalent to maximising the variance of the projected data. Why does this make PCA find the “directions of maximal variance”?

Question 2 – Choosing the number of components L

Several criteria were discussed for choosing the latent dimension L in PCA.

- What is a scree plot, and how can it be used to choose L ?
- Define the “fraction of variance explained” F_L . How does it behave as L increases, and how can it be used to choose L ?
- Briefly describe the idea behind the profile-likelihood method for choosing L . What quantity is being modelled, and what is being optimised?

Question 3 – From PCA to Factor Analysis

Factor analysis (FA) generalises PCA by introducing a probabilistic generative model.

- Write down the generative model for FA (the distributions for $\mathbf{z}$ and $\mathbf{x} | \mathbf{z}$), and the resulting marginal distribution $p(\mathbf{x})$.
- What role does the matrix Ψ play, and why must it be restricted to be diagonal?
- Compare the latent representation obtained from FA, $p(\mathbf{z} | \mathbf{x})$, with the latent representation obtained from PCA. What is the key conceptual difference?

Question 4 – Identifiability in Factor Analysis

The parameters of an FA model are not uniquely identifiable.

- Show (in words or with the relevant equation) why replacing $\mathbf{W}$ with $\tilde{\mathbf{W}} = \mathbf{W}\mathbf{R}$, for an orthogonal matrix $\mathbf{R}$, does not change the likelihood of the model.
- Give two of the proposed solutions to this identifiability problem, and briefly explain how each one resolves the ambiguity.

- c) Which of these solutions is used by PCA, and what additional ambiguity remains even after applying it?

Question 5 – Probabilistic PCA (PPCA)

PPCA is a special case of factor analysis.

- a) What additional restrictions does PPCA place on $\mathbf{W}$ and $\mathbf{\Psi}$, compared to general FA?
- b) In the noise-free limit $\sigma^2 \rightarrow 0$, how does the MLE $\mathbf{W}_{\text{mle}}$ relate to the PCA solution?
- c) What is the interpretation of the MLE for σ^2 in terms of the discarded eigenvalues?
- d) Write down the E step and M step of the EM algorithm for PPCA in the noise-free limit ($\sigma^2 = 0$).
- e) List two advantages of EM over standard eigenvector methods for fitting PCA.

Question 6 – Manifold learning and MDS

- a) What is the manifold hypothesis, and how does the digit-6 example illustrate the difference between intrinsic and ambient dimensionality?
- b) What is the goal of multidimensional scaling (MDS), and what quantity does classical MDS try to preserve?
- c) What is the difference between metric MDS and non-metric MDS? In particular, what does non-metric MDS try to match instead of distances?

Question 7 – t-SNE and Kernel PCA

- a) In SNE/t-SNE, what do the quantities $p_{j|i}$ and $q_{j|i}$ represent, and what objective is used to match them?
- b) What is the “crowding problem”, and how does t-SNE address it?
- c) What is the key idea behind kernel PCA, and what types of nonlinear structure can it capture that linear PCA cannot?